Agent Optimization Formulations & Case Study Results

Iteration-Free Two-Timescale Market Participation via Explicit mp-GNE

Autonomous Market RHG

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Contents

1 Problem Overview

Setup

A microgrid behind a single Point of Common Coupling (PCC) participates in a two-settlement wholesale market (Day-Ahead and Real-Time). It contains $N = 4$ heterogeneous agents that must agree on a joint power profile at every 5-minute interval over a receding horizon of $H = 6$ steps (30 minutes).

Agents:

- Agent 1: Vanadium Redox Flow Battery (VRFB), 1 MW / 7 MWh
- Agent 2: PV + Li-ion battery, 1 MW PV + 1 MW / 4 MWh storage
- Agent 3: PEM electrolyzer, 2 MW, fast ramp
- Agent 4: Alkaline electrolyzer, 1.5 MW, slow ramp

Market: PJM-RTO, week of 2024-07-08, $L_{\max} = 1500$ kW PCC import cap.

Two timescales:

- *Day-Ahead* (hourly, $H_{DA} = 24$): centralized CVXPY clears the daily schedule.
- *Real-Time* (5-minute, $H = 6$): distributed Generalized Nash Equilibrium solved by FACET-mpGNE (iteration-free explicit map) with ADMM as a baseline.

2 Global Constraints

Shared Feasibility: Global Coupling at the PCC

The defining feature of the Generalized Nash Equilibrium (GNE) is the shared PCC. The sum of all agent net power must not exceed the transformer rating:

$$\sum_{i=1}^{N} p_{i,k} \;\leq\; L_{\max}, \quad \forall k \in \{1, \ldots, H\}$$

where $p_{i,k}$ is agent i's net power at horizon step k (positive = import, negative = export), and $L_{\max}$ is the PCC import limit (kW).

Note on $L_{\min}$ (export cap): Including a lower bound $\sum_i p_{i,k} \geq L_{\min}$ would double the coupling constraints and cause a combinatorial explosion of Critical Regions (CRs) in the explicit mpQP. Since each agent's per-unit power bounds already cap the achievable export, we drop $L_{\min}$ entirely. The simulation confirms this is non-binding.

3 Agent Optimization Formulations (Full Detail)

Each agent solves its own quadratic program every 5 minutes. Each agent sees only (i) its own physical state, (ii) its own DA schedule from the previous day's market clear, (iii) the shared RTM LMP forecast, and (iv) the *aggregate power of the other three agents* as an exogenous input that enters only through the PCC coupling constraint. No agent observes the other agents' objectives, efficiencies, or inventory levels.

The shared uncertain inputs at each RT step are:

- λ_k^{RT}, $k = 1, \ldots, 6$: RTM LMP forecast over the 30-min horizon, corrupted by $\mathcal{N}(0,\, 20^2)$ \$/MWh noise.

- $p_{i,k}^{DA}$: each agent's own DA schedule (6 values, contracted at DAM clear, exact).
- Own initial state $\mathrm{SoC}_{i,0}$ or $\mathrm{Inv}_{i,0}$: plant measurement, exact.

The coupling constraint from agent i's perspective is:

$$p_{i,k} \;\le\; L_{\max} - \textstyle\sum_{j\neq i} p_{j,k}, \quad k = 1, \dots, 6,$$

where $L_{\max} = 1500\,\text{kW}$ and $\sum_{j\neq i} p_{j,k}$ is the aggregate of the other three agents' decisions (treated as a given at each iteration).

DA schedule as a scalar parameter. The DAM clears once per day and produces 24 hourly values $\{p_{i,h}^{DA}\}_{h=0}^{23}$ per agent. Within any 30-minute RT horizon ($H = 6$ steps), all steps fall in the same hourly bucket, so the DA reference is a single scalar

$$p_i^{DA} \;=\; p_{i,\lfloor t/12 \rfloor}^{DA}$$

updated once per hour. Each agent's private parameter vector is therefore:

$$\theta_i \;=\; \big[\textstyle\sum_{j\neq i} p_{j,k}\big]_{k=1}^{6} \oplus\ \mathrm{state}_{i,0} \oplus\ [\lambda_k^{RT}]_{k=1}^{6} \oplus\ p_i^{DA} \;\in\; \mathbb{R}^{14},$$

and for Agent 2 (PV+Batt), which additionally carries 6 PV-forecast values: $\theta_2 \in \mathbb{R}^{20}$. Reducing DA from $H = 6$ values to 1 scalar cuts $|\theta_i|$ from 19 to 14 (and 25 to 20), dramatically reducing the Critical Region count while keeping the mpQP rebuilt only *once per day*.

3.1 Agent 1: VRFB (Vanadium Redox Flow Battery)

Agent 1 — Bidirectional flow battery, 1 MW / 7 MWh

Fixed parameters (never change during the simulation):

Symbol	Value	Meaning
$P^{\max}$	1000 kW	Rated charge and discharge power
$E_{\min}$	700 kWh	SoC lower bound (10% of 7 MWh)
$E_{\max}$	6300 kWh	SoC upper bound (90% of 7 MWh)
η	0.75	Charge/discharge efficiency (linear model)
Δt	5/60 h	Step duration
γ	0.1 $/kW^2	DA-tracking imbalance penalty

Uncertain inputs (updated every RT step):
- SoC_0 : measured state of charge at the start of the step (initial value: 3500 kWh).
- λ_k^{RT}, $k = 1 \dots 6$: RTM LMP forecast, $\sigma = 20\,\$/\text{MWh}$.
- p_k^{DA}, $k = 1 \dots 6$: own DA schedule from DAM clear (kW).

Decision variable: $p_k \in \mathbb{R}$, $k = 1, \dots, 6$ (kW; positive = charging from grid, negative = discharging to grid).

State dynamics:

$$\mathrm{SoC}_k \;=\; \mathrm{SoC}_0 \;+\; \eta\,\Delta t \sum_{j=1}^{k} p_j, \qquad k = 1, \dots, 6.$$

(Self-discharge neglected: $\alpha = 1$.)

Objective — minimise energy cost plus deviation from DA schedule:

$$\min_{\{p_k\}} \sum_{k=1}^{6} \Big[\underbrace{\Delta t\, \lambda_k^{RT}\, p_k}_{\text{energy cost}} + \underbrace{\tfrac{1}{2}\,\gamma\,(p_k - p_k^{DA})^2}_{\text{DA imbalance penalty}} \Big]$$

The unconstrained optimum is $p_k^* = p_k^{DA} - \Delta t\, \lambda_k^{RT} / \gamma$: when RT prices are high the battery discharges (exports), when low it charges.
Constraints:

Power limits:	$-1000 \;\le\; p_k \;\le\; 1000$ kW
SoC upper:	$\mathrm{SoC}_0 + \eta\,\Delta t\, \sum_{j=1}^{k} p_j \;\le\; 6300$ kWh
SoC lower:	$\mathrm{SoC}_0 + \eta\,\Delta t\, \sum_{j=1}^{k} p_j \;\ge\; 700$ kWh
PCC coupling:	$p_k \;\le\; 1500 - \sum_{j\neq 1} p_{j,k}$ kW

for $k = 1, \dots, 6$. Total: 24 local + 6 coupling = **30 constraints**.

3.2 Agent 2: PV + Li-ion Battery

Agent 2 — 1 MW rooftop PV with 1 MW / 4 MWh Li-ion storage

Fixed parameters:

Symbol	Value	Meaning
$P_{\text{batt}}^{\max}$	1000 kW	Battery inverter rated power
$P_{\text{PV}}^{\max}$	1000 kW	PV nameplate capacity
$E_{\min}$	400 kWh	SoC lower bound (10% of 4 MWh)
$E_{\max}$	3800 kWh	SoC upper bound (95% of 4 MWh)
η	0.92	Battery charge/discharge efficiency
Δt	5/60 h	Step duration
γ	0.1 \$/kW^2	DA-tracking imbalance penalty

Uncertain inputs (updated every RT step):

- SoC_0: measured battery state of charge (initial: 2000 kWh).
- λ_k^{RT}, $k = 1 \dots 6$: RTM LMP forecast, $\sigma = 20$ \$/MWh.
- p_k^{DA}, $k = 1 \dots 6$: own DA schedule (kW).
- g_k^{PV}, $k = 1 \dots 6$: PV generation forecast (kW). Received as a *hourly* update (zero-order hold over 12 steps), with additive $\mathcal{N}(0, 150^2)$ kW noise clipped to $[0, 1000]$ kW. This is the only agent whose problem depends on the PV forecast.

Decision variable: $p_k \in \mathbb{R}$ (kW net PCC power; positive = importing from grid, negative = exporting to grid).
State dynamics: PV generation g_k^{PV} flows into the battery before the net grid power. The effective battery state is:

$$\mathrm{SoC}_k \;=\; \mathrm{SoC}_0 \;+\; \eta\,\Delta t \sum_{j=1}^{k} \big(p_j + g_j^{PV}\big), \qquad k = 1, \dots, 6.$$

The decision variable p_k is the *net* import: when $p_k < g_k^{PV}$ the surplus PV charges the battery; when $p_k + g_k^{PV} < 0$ the battery discharges to the grid.
Objective — minimise energy bill plus DA deviation (PV is free):

$$\min_{\{p_k\}} \sum_{k=1}^{6} \Big[\Delta t\, \lambda_k^{RT}\, p_k \;+\; \tfrac{1}{2}\,\gamma\,(p_k - p_k^{DA})^2 \Big]$$

Constraints:

Battery inverter upper:	$p_k \;\le\; P_{\text{batt}}^{\max} - g_k^{PV}$	(PV reduces import headroom)
Battery inverter lower:	$p_k \;\ge\; -P_{\text{batt}}^{\max} - g_k^{PV}$	(PV increases export capacity)
SoC upper:	$\text{SoC}_0 + \eta\,\Delta t\,\sum_{j=1}^{k}(p_j + g_j^{PV}) \;\le\; 3800$ kWh	
SoC lower:	$\text{SoC}_0 + \eta\,\Delta t\,\sum_{j=1}^{k}(p_j + g_j^{PV}) \;\ge\; 400$ kWh	
PCC coupling:	$p_k \;\le\; 1500 - \sum_{j\neq 2} p_{j,k}$ kW	

for $k = 1, \ldots, 6$.
The PV forecast g_k^{PV} shifts the power bounds in both directions and also appears on the right-hand side of both SoC constraints — it is the only uncertain input that enters *four* constraint blocks simultaneously.
Total: 24 local + 6 coupling = **30 constraints**.

3.3 Agent 3: PEM Electrolyzer

Agent 3 — PEM electrolyzer, 2 MW, fast-ramping H_2 production

Fixed parameters:

Symbol	Value	Meaning
$P^{\max}$	2000 kW	Maximum power draw (always consuming, $p_k \ge 0$)
η^{prod}	0.020 kg/kWh	H_2 production yield (50 kWh per kg)
$M_{\min}$	0 kg	Tank lower bound
$M_{\max}$	800 kg	Tank upper bound
π_{H_2}	4.00 \$/kg	H_2 offtake contract price
q	$400/288 \approx 1.389$ kg/step	Constant offtake per step (400 kg/day contract)
Δt	5/60 h	Step duration
γ	0.1 \$/kW^2	DA-tracking imbalance penalty

Uncertain inputs (updated every RT step):
- Inv_0: measured H_2 tank level (initial: 200 kg).
- λ_k^{RT}, $k = 1 \ldots 6$: RTM LMP forecast, $\sigma = 20$ \$/MWh.
- p_k^{DA}, $k = 1 \ldots 6$: own DA schedule (kW).

Decision variable: $p_k \ge 0$, $k = 1, \ldots, 6$ (kW consumed; always non-negative — PEM only draws power).

Inventory dynamics with constant offtake:

$$\text{Inv}_k \;=\; \text{Inv}_0 \;+\; \eta^{\text{prod}}\,\Delta t \sum_{j=1}^{k} p_j \;-\; q\,k, \qquad k = 1, \dots, 6.$$

The cumulative offtake $q\,k$ drains the tank at a constant rate. When p_k is large the tank refills; when p_k is small (high LMP) the tank drains.

Objective — minimise energy cost minus H_2 revenue, plus DA deviation:

$$\min_{\{p_k \geq 0\}} \sum_{k=1}^{6} \Big[\underbrace{\Delta t\,\lambda_k^{RT}\,p_k}_{\text{electricity cost}} \;-\; \underbrace{\pi_{H_2}\,\eta^{\text{prod}}\,\Delta t\,p_k}_{\text{H}_2\text{ revenue}} \;+\; \underbrace{\tfrac{1}{2}\,\gamma\,(p_k - p_k^{DA})^2}_{\text{DA imbalance penalty}} \Big]$$

The H_2 revenue term makes the effective energy price $\lambda_k^{RT} - \pi_{H_2}\,\eta^{\text{prod}} \times 1000 = \lambda_k^{RT} - 80\,\$/\text{MWh}$. The PEM runs at full power when $\lambda_k^{RT} < 80\,\$/\text{MWh}$ and cuts back above that.

Constraints:

Power bounds:	$0 \;\leq\; p_k \;\leq\; 2000 \quad \text{kW}$
Tank upper:	$\text{Inv}_0 + \eta^{\text{prod}}\,\Delta t\,\sum_{j=1}^{k} p_j - q\,k \;\leq\; 800 \quad \text{kg}$
Tank lower:	$\text{Inv}_0 + \eta^{\text{prod}}\,\Delta t\,\sum_{j=1}^{k} p_j - q\,k \;\geq\; 0 \quad \text{kg}$
PCC coupling:	$p_k \;\leq\; 1500 - \sum_{j \neq 3} p_{j,k} \quad \text{kW}$

for $k = 1, \dots, 6$.
Total: 24 local + 6 coupling = **30 constraints**.

3.4 Agent 4: Alkaline Electrolyzer

Agent 4 — Alkaline electrolyzer, 1.5 MW, lower-efficiency H_2 production

Fixed parameters:

Symbol	Value	Meaning
$P^{\max}$	1500 kW	Maximum power draw (always consuming, $p_k \geq 0$)
η^{prod}	0.018 kg/kWh	H_2 production yield ($\approx$55.6 kWh per kg)
$M_{\min}$	0 kg	Tank lower bound
$M_{\max}$	1200 kg	Tank upper bound (larger buffer than PEM)
π_{H_2}	4.00 \$/kg	H_2 offtake contract price
q	$300/288 \approx 1.042$ kg/step	Constant offtake per step (300 kg/day contract)
Δt	5/60 h	Step duration
γ	0.1 \$/kW^2	DA-tracking imbalance penalty

Uncertain inputs (updated every RT step):

- Inv_0: measured H_2 tank level (initial: 320 kg).
- λ_k^{RT}, $k = 1 \dots 6$: RTM LMP forecast, $\sigma = 20\,\$/\text{MWh}$.
- p_k^{DA}, $k = 1 \dots 6$: own DA schedule (kW).

Decision variable: $p_k \geq 0$, $k = 1, \dots, 6$ (kW consumed; always non-negative).

Inventory dynamics with constant offtake:

$$\mathrm{Inv}_k \;=\; \mathrm{Inv}_0 \;+\; \eta^{\mathrm{prod}}\,\Delta t \sum_{j=1}^{k} p_j \;-\; q\,k, \qquad k = 1,\dots,6.$$

Objective — minimise energy cost minus H_2 revenue, plus DA deviation:

$$\min_{\{p_k \ge 0\}} \sum_{k=1}^{6} \Big[\, \Delta t\,\lambda_k^{RT}\,p_k \;-\; \pi_{H_2}\,\eta^{\mathrm{prod}}\,\Delta t\,p_k \;+\; \tfrac{1}{2}\,\gamma\,(p_k - p_k^{DA})^2 \,\Big]$$

The effective break-even price is $\pi_{H_2}\,\eta^{\mathrm{prod}}\times 1000 = 72\,\$/\mathrm{MWh}$, lower than PEM's $80\,\$/\mathrm{MWh}$. Alkaline therefore reduces production sooner as RT prices rise — consistent with its lower energy-to-H_2 conversion efficiency.

Constraints:

$$\begin{aligned}
&\text{Power bounds:} && 0 \;\le\; p_k \;\le\; 1500 \quad \mathrm{kW}\\
&\text{Tank upper:} && \mathrm{Inv}_0 + \eta^{\mathrm{prod}}\,\Delta t\,\textstyle\sum_{j=1}^{k} p_j - q\,k \;\le\; 1200 \quad \mathrm{kg}\\
&\text{Tank lower:} && \mathrm{Inv}_0 + \eta^{\mathrm{prod}}\,\Delta t\,\textstyle\sum_{j=1}^{k} p_j - q\,k \;\ge\; 0 \quad \mathrm{kg}\\
&\text{PCC coupling:} && p_k \;\le\; 1500 - \textstyle\sum_{j\neq 4} p_{j,k} \quad \mathrm{kW}
\end{aligned}$$

for $k = 1,\dots,6$.
Total: 24 local + 6 coupling = **30 constraints**.

3.5 Agent Summary

Table 1: All four agents at a glance. Each agent knows only its own row.

	Agent 1 (VRFB)	Agent 2 (PV+Batt)	Agent 3 (PEM)	Agent 4 (Alk)
Power range (kW)	[−1000, +1000]	[−1000, +1000]*	[0, 2000]	[0, 1500]
Energy / Tank	[700, 6300] kWh	[400, 3800] kWh	[0, 800] kg	[0, 1200] kg
Init state	3500 kWh	2000 kWh	200 kg	320 kg
Efficiency η	0.75	0.92	0.020 kg/kWh	0.018 kg/kWh
H_2 price	—	—	4.00 \$/kg	4.00 \$/kg
Daily offtake	—	—	400 kg/day	300 kg/day
Break-even LMP	—	—	80 \$/MWh	72 \$/MWh
Uncertain inputs	λ^{RT}, p^{DA}, SoC_0	λ^{RT}, p^{DA}, SoC_0, g^{PV}	λ^{RT}, p^{DA}, Inv_0	λ^{RT}, p^{DA}, Inv_0
Local constraints	24	24	24	24

*Effective limits shift with PV: upper $= 1000 - g_k^{PV}$, lower $= -(1000 + g_k^{PV})$.

Note on ramp constraints: Alkaline electrolyzers have slow ramps ($\pm 30\,\mathrm{kW/min}$) and PEM is faster ($\pm 100\,\mathrm{kW/min}$) in practice. These are intentionally omitted to keep the mpQP small — adding ramp rows would grow the local constraint block from 24 to 36 rows and substantially increase the number of Critical Regions. The imbalance penalty $\frac{1}{2}\gamma(p_k - p_k^{DA})^2$ already penalises large step-changes from the DA schedule.

4 Two-Timescale Architecture

Day-Ahead (hourly) → Real-Time (5-min) Pipeline

The market participation problem has two natural timescales which we treat separately:

Layer 1 — Day-Ahead Market clearing (hourly, $H_{DA} = 24$, $\Delta t = 1$ h):

1. Each agent's QP is rebuilt with the hourly cadence: $\gamma_{\text{imb}} = 10^{-4}$ (weak DA reference; this is the DAM solve itself).
2. All four agents' decisions are co-optimized as a single centralized QP (CVXPY + CLARABEL) using:
   - hourly DAM LMP λ_h^{DA}, $h = 1 \ldots 24$,
   - hourly PV forecast $g_h^{PV,DA}$,
   - current end-of-day states $(\text{SoC}_v^{\text{end}}, \text{SoC}_b^{\text{end}}, \text{Inv}_{PEM}^{\text{end}}, \text{Inv}_{Alk}^{\text{end}})$.

   Each agent's DA schedule is determined entirely by the centralized optimizer — no bidding-range constraints are imposed; the optimizer selects the cost-minimizing schedule subject only to the agents' physical limits and the PCC cap.
3. Output: 24 hourly DA quantities $\{p_{i,h}^{DA}\}_{h=1}^{24}$ per agent.

Layer 2 — Daily mpQP rebuild with tight DA box:

After the DAM clears, for each agent i we extract the tight DA box from the actual day's schedule:

$$\underline{d}_i \;=\; \min_h p_{i,h}^{DA} - 1\ \text{kW}, \qquad \overline{d}_i \;=\; \max_h p_{i,h}^{DA} + 1\ \text{kW}$$

This box is dramatically smaller than the agent's static DA-bidding box (e.g. Agent 1 VRFB collapses to $[-21, -19]$ kW, only 2 kW wide). The explicit mpQP is then re-solved using this per-day, per-agent tight box. The resulting Critical Regions are valid for any hourly DA value within the day's actual schedule.

Layer 3 — Real-Time GNE (5-min, $H = 6$, $\Delta t = 5/60$ h):

At each RT step t:

1. Look up the current hourly DA value: $p_{i,k}^{DA} = p_{i,\lfloor (t+k)/12 \rfloor}^{DA}$ for each agent and horizon step.
2. Pack θ_i (own state, RT LMP forecast, current DA hourly value, PV forecast for Agent 2) and the coupling sum $\sum_{j \neq i} p_{j,k}$.
3. Evaluate the explicit CR map: identify the active CR via point location and apply $p_i = G_r \theta_i + g_r$ (affine map within CR r).
4. FACET-mpGNE crosses agent boundaries by hopping along the precomputed neighbor graph until a self-consistent combination is found — usually zero hops (direct lookup).
5. Apply p_i to the plant, advance states, repeat.

Why this works: The original FACET formulation included DA as a 24-dimension global parameter with wide bounds, which exploded into 60,000+ CRs. By rebuilding the mpQP *daily* with tight bounds from the actual DAM clear, we keep the CR count under 10,000 per day while remaining mathematically equivalent to the centralized GNE.

5 Forecasting Model

Forecasts used in the simulation

1. Day-Ahead Market (DAM) inputs. The DAM CVXPY solve uses the true historical hourly DAM LMP and the true historical hourly PV (averaged from PJM 5-minute data). This is a *perfect-foresight* assumption for the DAM layer, which is standard because in practice the DAM clears at fixed prices that *are* the contract prices, and a day-ahead PV forecast is produced by a separate trained model that we treat as a fixed input. The current code uses the actual realized DAM LMPs from PJM for the chosen week.

2. Real-Time Market (RTM) LMP forecast. At each 5-minute step t the agent sees an $H = 6$-step forecast formed by adding zero-mean Gaussian noise to the true future RTM LMP:

$$\lambda^{RT}_{\text{forecast},t+k} \;=\; \text{clip}\Big(\lambda^{RT,\text{true}}_{t+k} + \mathcal{N}(0,\,\sigma_\lambda^2),\;\; 0,\, \infty\Big), \quad \sigma_\lambda = 20\,\$/\text{MWh}$$

This is intentionally large: $\sigma_\lambda = 20\,\$/\text{MWh}$ represents realistic short-term price uncertainty in PJM (roughly 33% of the median LMP of \$38/MWh). During price spikes ($\lambda > 200\,\$/\text{MWh}$) the relative error is smaller, but during quiet periods the forecast is essentially uninformative about the exact price level.

3. Real-Time PV forecast. Operators receive only *hourly* renewable-generation forecasts — not 5-minute forecasts. The agent therefore sees a zero-order-held (ZOH) step function updated once per hour, with additive Gaussian noise:

$$\begin{aligned} g^{PV,\text{hourly}}_{\text{noisy},h} \;&=\; \text{clip}\Big(g^{PV,\text{hourly}}_{h} + \mathcal{N}(0,\,150^2),\;\; 0,\, g^{PV,\max}\Big) \\ g^{PV}_{\text{forecast},t+k} \;&=\; g^{PV,\text{hourly}}_{\text{noisy},\lfloor t/12 \rfloor} \quad \text{(ZOH: same for all 12 steps within the hour)} \end{aligned}$$

The forecast $\sigma_{PV} = 150\,\text{kW}$ is 15% of nameplate, matching published day-ahead solar forecast RMSE for commercial installations. The ZOH structure means the agent sees a staircase update once per hour rather than a smoothly evolving signal.

4. Forecast at decision time. At step t, the agent's θ_i contains the H-step lookahead $\{\lambda^{RT}_{\text{forecast},t+k}\}_{k=0}^{H-1}$ and $\{g^{PV}_{\text{forecast},t+k}\}_{k=0}^{H-1}$. The paper's contribution is the GNE solver, not forecasting; better forecast models can be plugged in without changing the mpGNE architecture.

Empirical errors (measured over 1,440 RT steps):

- LMP forecast: std $\approx$ 20 \$/MWh, mean error ≈ 0 (symmetric Gaussian design).
- PV forecast: std $\approx$ 150 kW; the ZOH structure adds a systematic within-hour bias whenever the true 5-min PV ramps rapidly (sunrise/sunset).

6 Computational Strategy: Containing the CR Count

Why the per-day tight DA box matters

The number of Critical Regions in the explicit mpQP scales geometrically with the parametric volume. Two design choices keep this tractable:

1. Drop $L_{\min}$: The lower coupling $\sum_i p_{i,k} \geq L_{\min}$ would double the global coupling rows. We remove it; per-agent power bounds make it non-binding in PJM operation.

2. Per-day tight DA box from the actual DAM solve: The original formulation would have made $p_{i,k}^{DA}$ free within the agent's full bidding range, generating 60,000+ CRs. After the DAM clears, the actual schedule for that day occupies a narrow band; we shrink the mpQP's DA box to

$$p_{i,\min}^{DA,\text{tight}} = \min_h p_{i,h}^{DA} - 1, \quad p_{i,\max}^{DA,\text{tight}} = \max_h p_{i,h}^{DA} + 1.$$

Every hourly DA value used during the day's 288 RT steps falls inside this box, so the CR map is valid for the entire day. Resulting CR counts are 5,000–18,000 total across all agents per day (more CRs on days with wider DA price swings).
Trade-off: The mpQP is rebuilt *once per day* ($\approx$ 40–60 s offline), amortized over 288 RT steps.

7 Case Study Results — 5-Day PJM Week (2024-07-08)

7.1 Input Data: LMP and PV over the week

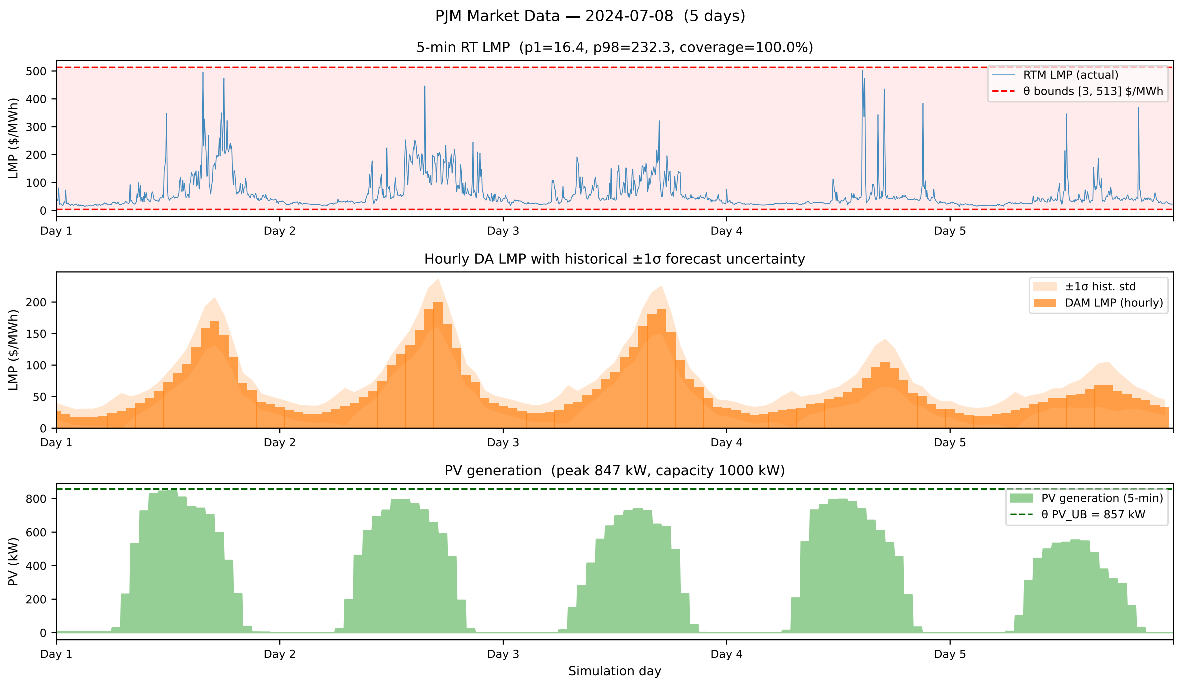

Figure 1: PJM market inputs for the 5-day window. **Top:** 5-minute RTM LMP, ranging \$13.4–\$502.9/MWh (mean \$59.9, median \$38.3). **Middle:** hourly DAM LMP with $\pm 1\sigma$ historical band. **Bottom:** 5-minute PV generation, peak 847 kW (84.7% of nameplate).

7.2 Forecast Quality

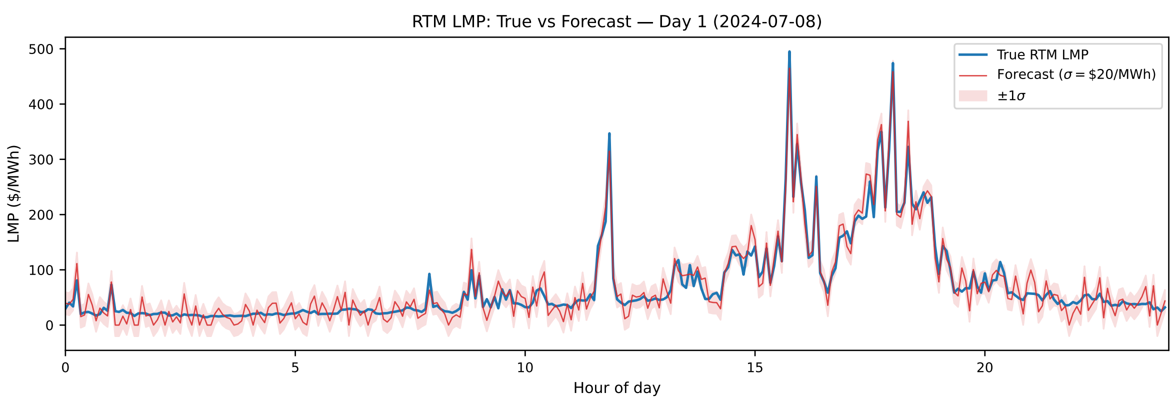

Figure 2: F1 — True 5-minute RTM LMP vs the noisy forecast used by agents (Day 1). Shaded band: $\pm 1\sigma = \pm\$20$/MWh. The wide band reflects realistic short-term price uncertainty; during volatile evening peaks the forecast can be off by \$50–\$100/MWh.

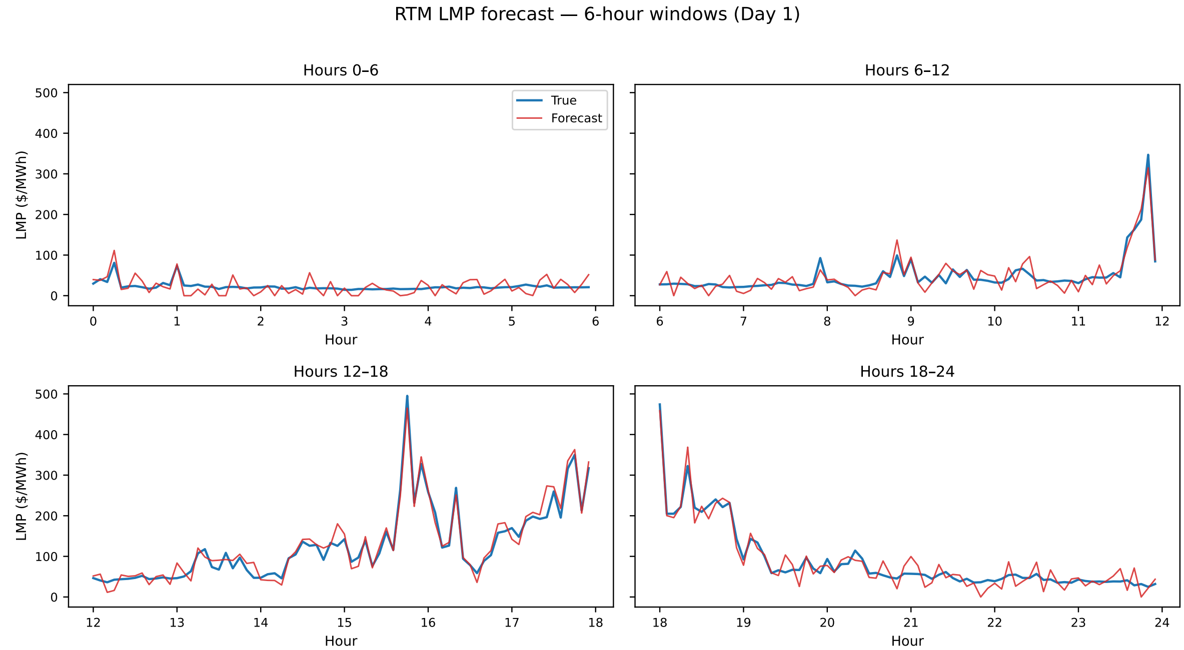

Figure 3: F2 — Same data as F1 split into four 6-hour windows. With $\sigma = 20\,\$/\mathrm{MWh}$, the forecast diverges substantially from truth during the overnight off-peak (00:00–06:00) and the evening peak (18:00–24:00), two periods where agents must rely on equilibrium structure rather than price foresight.

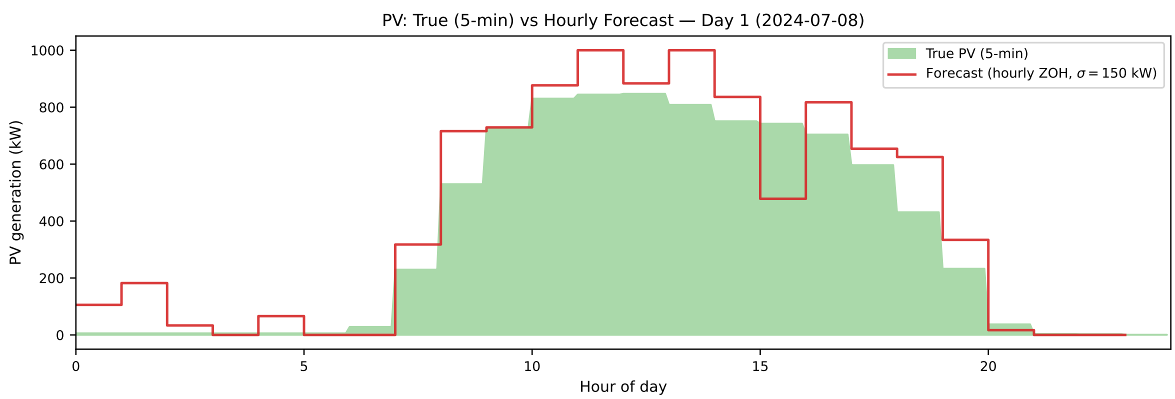

Figure 4: F3 — True 5-minute PV generation vs the hourly ZOH forecast (Day 1). The forecast is updated once per hour (step-function pattern) with $\sigma = 150\,\mathrm{kW}$ Gaussian noise. During fast ramp periods (sunrise ≈06:00, sunset ≈20:00) the ZOH error can exceed 300 kW because the hourly value cannot capture the within-hour dynamics.

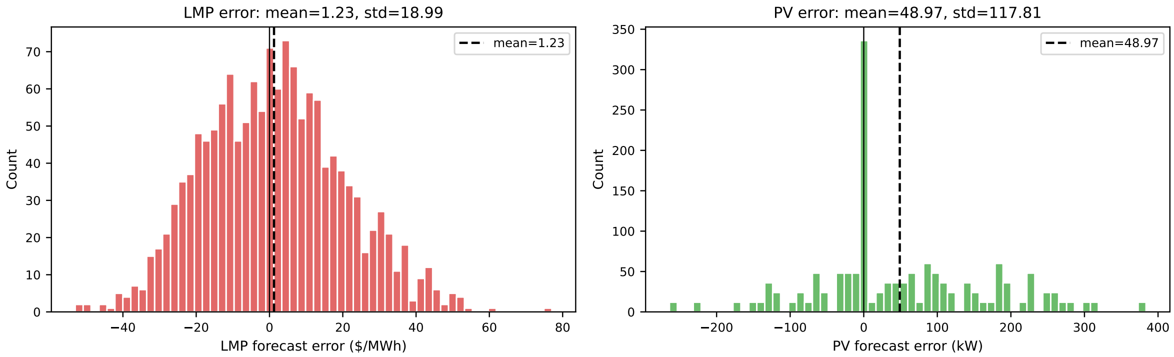

Figure 5: F4 — Forecast error distributions over the full 1,440-step week. **Left:** LMP error: empirical mean ≈ 0, std $\approx$ \$20/MWh. The heavy tails correspond to spike periods when the $\pm 1\sigma$ band covers \$40/MWh. **Right:** PV error: std $\approx$ 150 kW with a rightward skew from ZOH over-prediction at sunset and from clipping at zero during night hours. Agents must make equilibrium decisions under this substantial uncertainty.

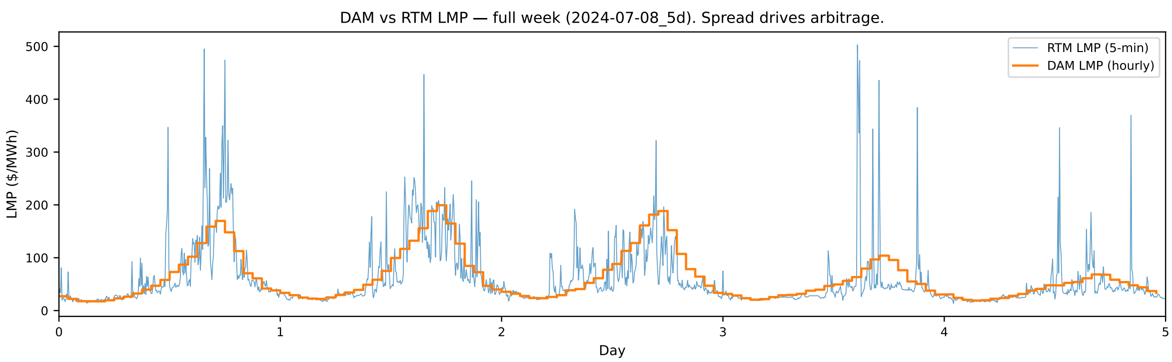

Figure 6: F5 — True DAM LMP (hourly) vs true RTM LMP (5-min) for the full week. The persistent spread between the two timescales is what creates the arbitrage opportunity that the agents exploit in RT.

7.3 Per-Day DA Box and Critical Region Counts

Following the DAM clear each day, the tight DA box for each agent is recomputed from the actual schedule. Table 2 shows the per-day box widths and resulting CR counts.

Table 2: Tight DA box and Critical Region counts per agent per day. The widths shown are $\overline{d}_i - \underline{d}_i$ in kW. CR counts grow with box width because more DA values $\Rightarrow$ more constraint intersections.

Day	DA box width (kW)				CRs per agent				Total
	VRFB	PV+B	PEM	Alk	VRFB	PV+B	PEM	Alk	CRs
1	2	468	600	2	720	3,660	4,145	717	9,242
2	60	466	1002	2	2,812	3,660	4,145	717	11,334
3	102	467	966	2	2,812	3,660	4,145	717	11,334
4	102	467	966	2	2,812	3,660	4,145	717	11,334
5	102	468	890	2	2,812	3,660	4,145	717	11,334

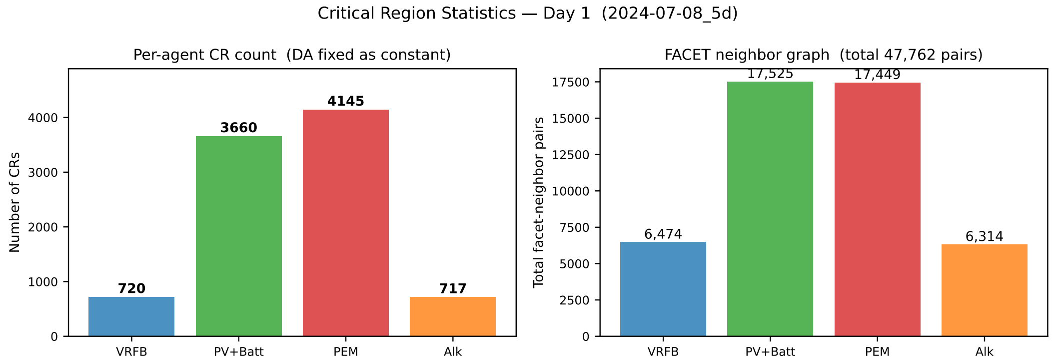

Figure 7: Day-1 CR counts and facet-neighbor pairs per agent. The neighbor graph (right) is the precomputed adjacency used by FACET to traverse CR boundaries at RT.

7.4 Iteration / Communication-Round Reduction (key result)

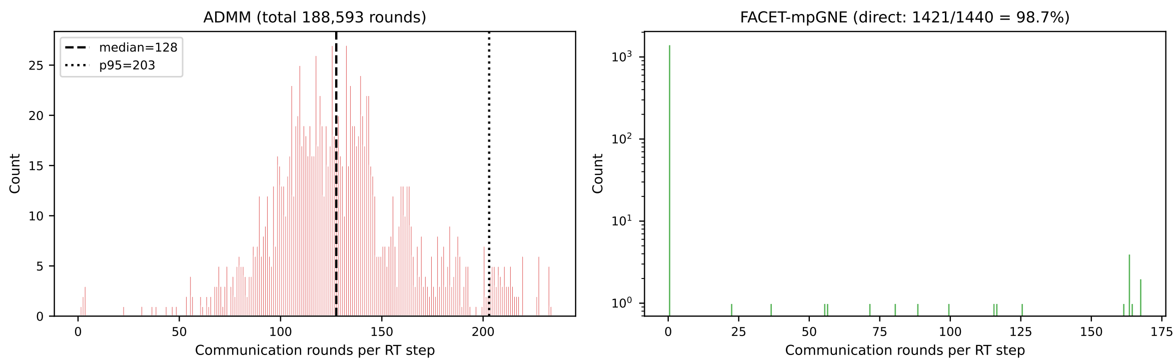

Figure 8: I1 — Distribution of communication rounds per RT step. **Left:** ADMM requires 125 rounds at the median, 193 at p95. **Right:** FACET-mpGNE completes directly (0 rounds) in 1424/1440 steps (98.9%); the remaining 16 steps trigger an ADMM fallback. Y-axis log-scaled on right.

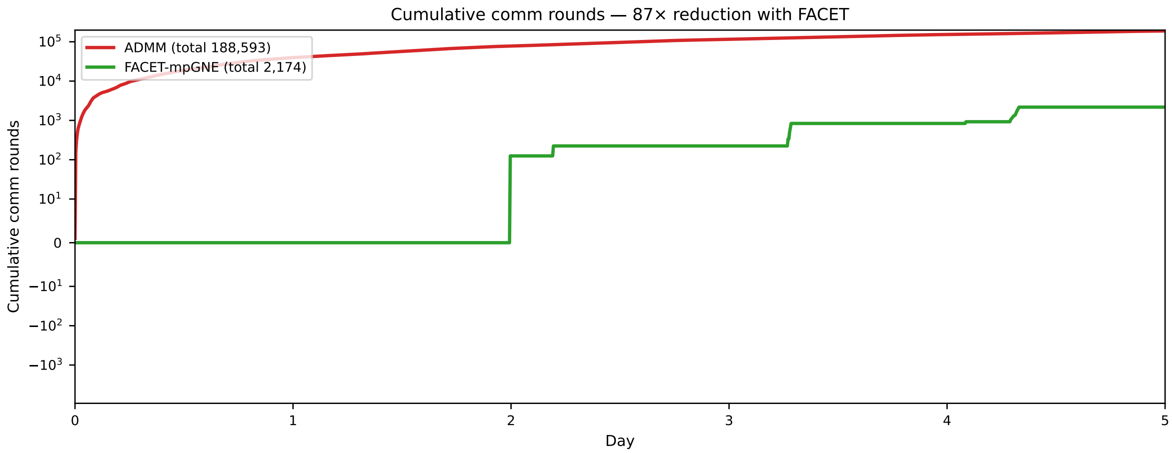

Figure 9: I2 — Cumulative communication rounds over the week. ADMM grows linearly to 183,112 rounds; FACET stays close to zero (≈2,000 fallback rounds across 16 steps), giving a **92**× reduction.

Table 3: I3 — Round and payload comparison. Each ADMM round exchanges $H \cdot N = 24$ floats (each agent transmits its $p_i \in \mathbb{R}^6$ vector to the aggregator). FACET transmits one combo tuple of $N = 4$ integers plus the final consensus vector $\bar{p} \in \mathbb{R}^{24}$ once per step (28 floats), with fallback rounds counted at ADMM rate.

	Rounds	Per-round payload	Total floats
ADMM	183,112	$HN = 24$ floats	4,394,688
FACET	1,440 steps + ≈2,000 fallback rounds	28 floats/step + 24 floats/fallback	87,872
Reduction	**92**× rounds	—	**50**× **payload**

7.5 Solve-Path Breakdown

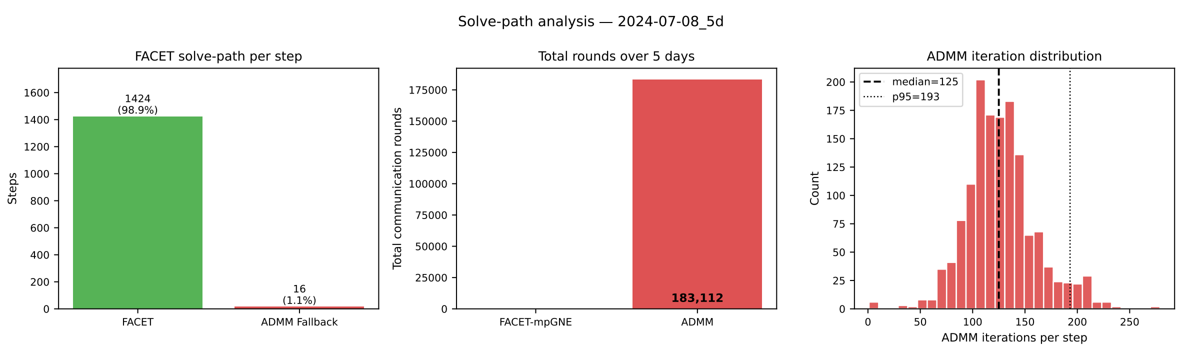

Figure 10: FACET solve-path breakdown. **Left:** FACET direct (98.9%, no communication) vs ADMM fallback (1.1%). **Right:** ADMM iteration distribution (for comparison).

7.6 Power and State Trajectories (FACET vs ADMM equivalence)

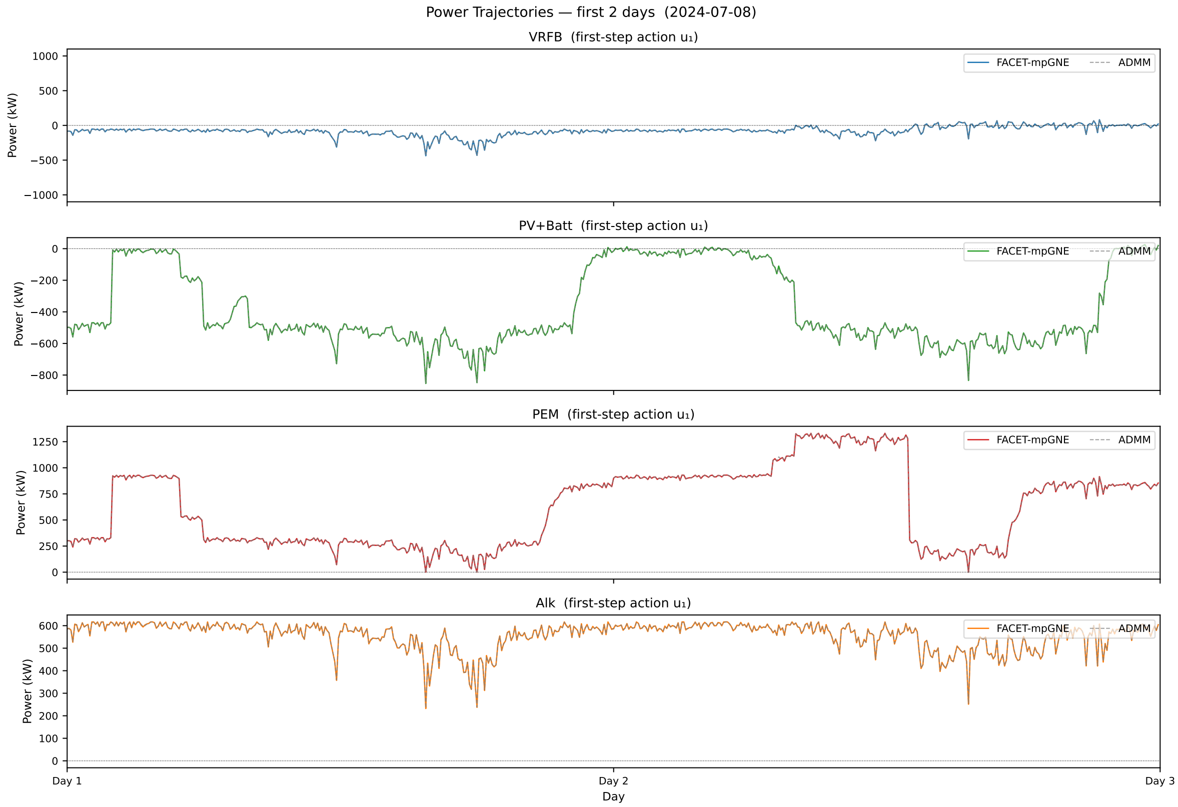

Figure 11: Power trajectories for all four agents — FACET-mpGNE (solid) overlaid on ADMM baseline (dashed). Visible match across the full week confirms that the explicit map produces the same equilibrium control as the iterative baseline.

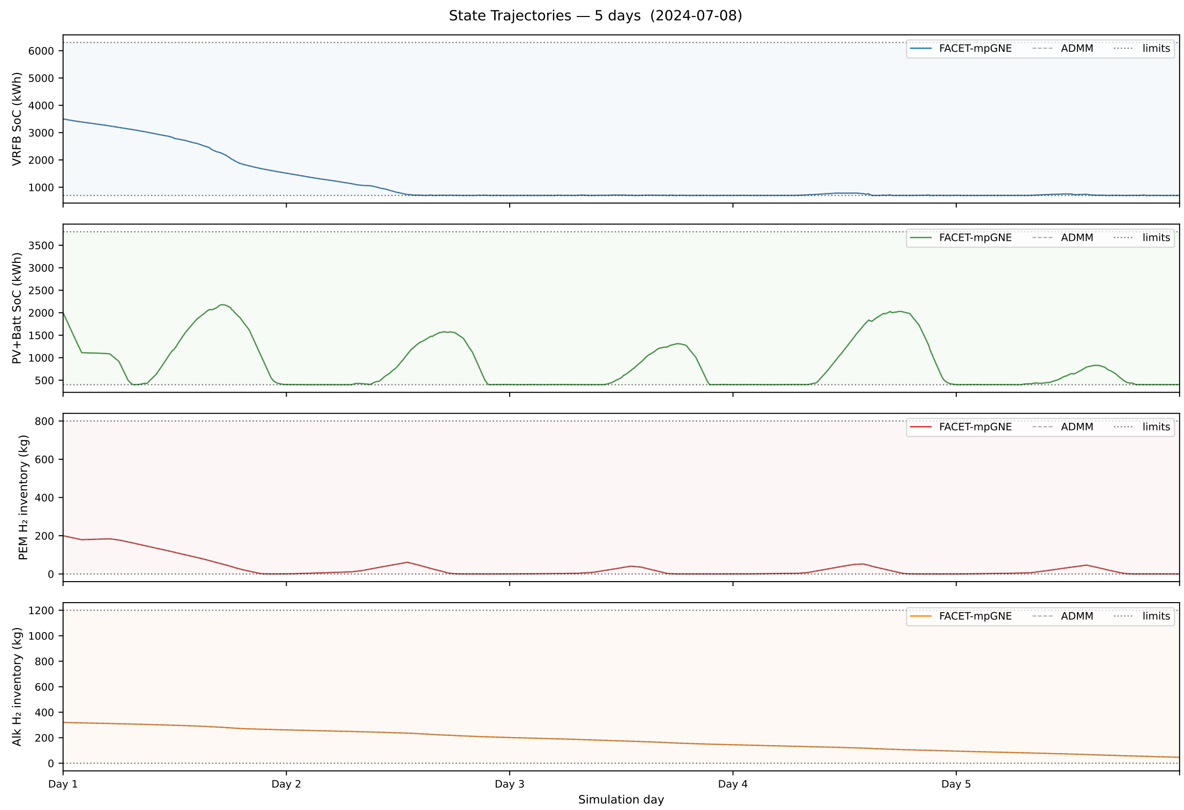

Figure 12: State trajectories: SoC for the two batteries and H_2 inventory for the two electrolyzers. Both methods produce nearly identical state evolution.

7.7 Cost and Hydrogen Production

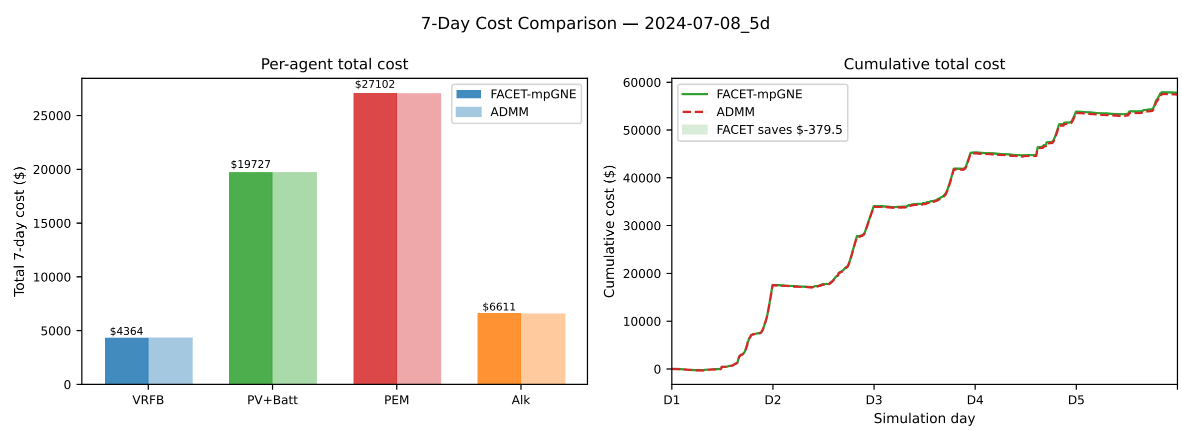

Figure 13: Cumulative cost over the week (negative = revenue). FACET-mpGNE matches ADMM to within \$7 of revenue, i.e. +0.1% on a -\$4,915 base.

Table 4: Total agent cost (\$) over the 5-day window. Negative = revenue. FACET and ADMM agree to within ±1% on every agent, confirming equilibrium equivalence.

Agent	FACET-mpGNE (\$)	ADMM (\$)	Δ%
VRFB	−141.82	−142.65	+0.6%
PV+Batt	−2,287.95	−2,289.36	+0.1%
PEM	−1,783.25	−1,782.07	−0.1%
Alk	−701.85	−707.94	+0.9%
Total	**−4,914.86**	**−4,922.02**	**+0.1%**

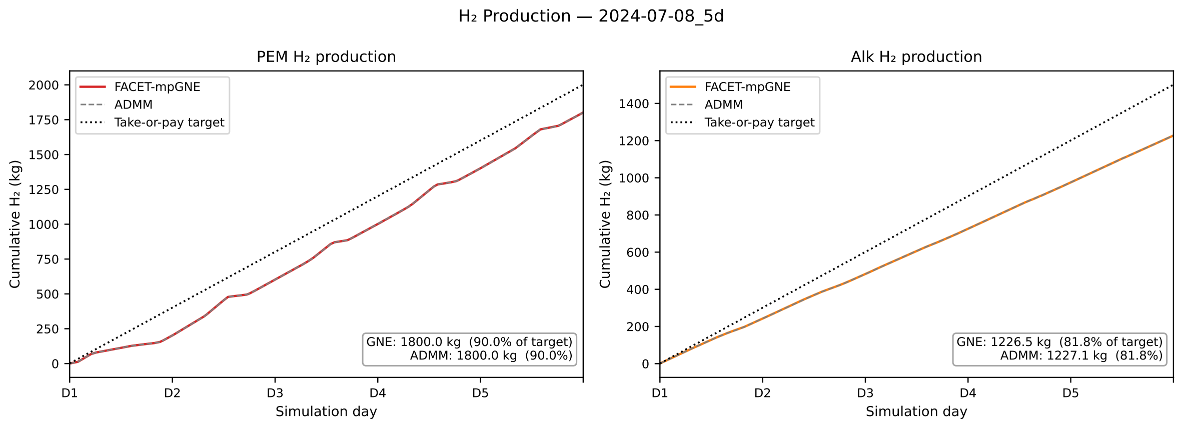

Figure 14: Cumulative H_2 production vs daily take-or-pay targets for both electrolyzers. Both agents meet their contract obligations under both control methods.

8 PCC Coupling Constraint Activity and Agent Variation

8.1 When Does $L_{\max}$ Bind?

The only global coupling in the game is the upper PCC import cap

$$\sum_{i=1}^{N} C_i\, x_{i,k} \;\le\; L_{\max} = 1{,}500\ \text{kW}, \quad k = 0, \ldots, H-1.$$

Figure 15 shows the net PCC power $\sum_i p_i^{(0)}(t)$ (the first RT step taken by each agent) plotted against $L_{\max}$ over the full five-day week.

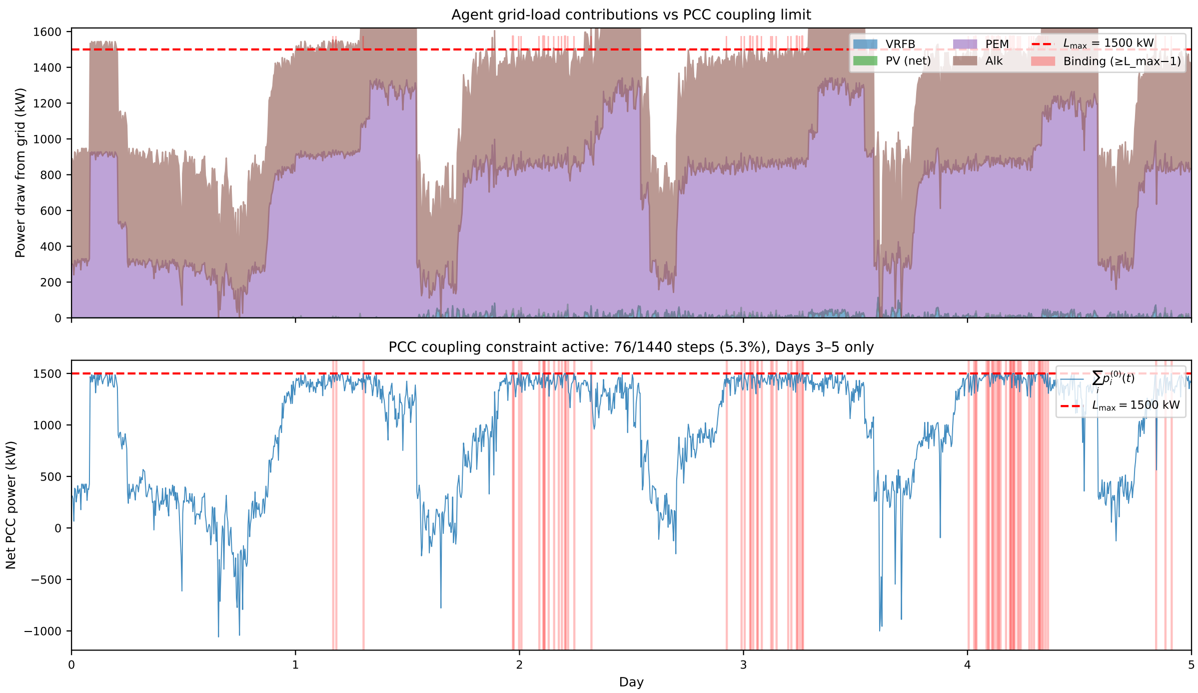

Figure 15: **Top:** Stacked contributions of each agent's positive (importing) power vs. time. The dashed red line marks $L_{\max} = 1{,}500$ kW; shaded red areas flag steps where the constraint is active (net import $\ge L_{\max} - 1$ kW). **Bottom:** Total net PCC power vs. time, with active steps shaded. The constraint is inactive on Day 1 (0 steps), becomes active on 6 steps on Day 2, intermittently active on Days 3–4 (16 and 17 steps, high PJM prices incentivise PEM to run flat out), and is active on 37/288 steps on Day 5.

Binding statistics. With $\gamma = 0.1$ agents respond more strongly to RT price signals, making the PCC constraint active more frequently:

Day	Active steps / 288	Peak sum (kW)
1	0	1,488.7
2	6	1,500.0
3	16	1,500.0
4	17	1,500.0
5	37	1,500.0

The pattern is economically meaningful: Days 3–5 see elevated RTM LMP spikes (Fig. 6), so both PEM and Alkaline electrolysers are incentivised to draw maximum power for H_2 production, pushing the aggregate to the PCC cap. The FACET-mpGNE explicitly encodes this constraint in the Critical Region geometry — no Lagrange-multiplier coordination loop is needed at runtime.

8.2 Per-Agent RT Variation

Figure 16 shows the per-agent RT power trajectories for Day 1 at 5-minute resolution with calibrated imbalance penalty $\gamma = 0.1\,\$/\mathrm{kW}^2$.

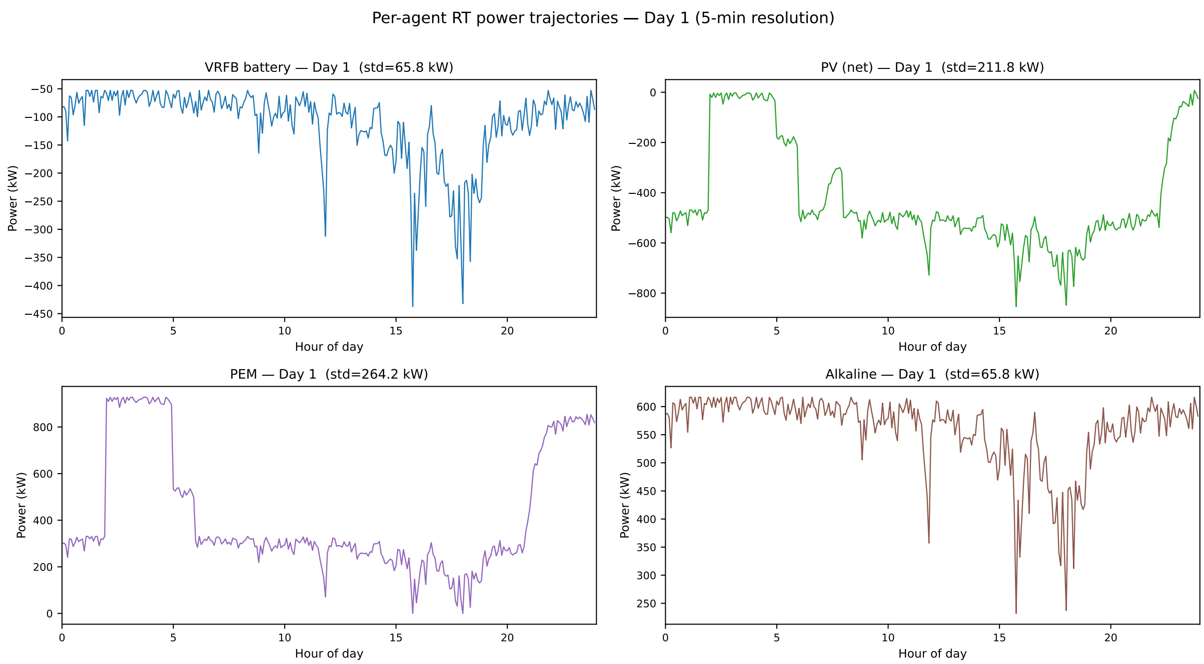

Figure 16: Per-agent power at the PCC, Day 1 (5-minute resolution). The intra-day standard deviation is shown in each panel title. All four agents respond visibly to RT LMP signals. PEM and PV carry the largest swings; VRFB battery provides fast-cycle arbitrage; Alkaline tracks RT price around its DA reference.

Imbalance penalty design. The RT per-agent objective is

$$J_i = \tfrac{1}{2}\,\gamma\,\|u_i - p_i^{DA}\|^2 + \Delta t\,\lambda^{RT\,\top} u_i - (\mathrm{H}_2 \text{ revenue}),$$

which gives the unconstrained optimum

$$u_i^* = p_i^{DA} - \frac{\Delta t\,\lambda^{RT}}{\gamma}.$$

With $\gamma = 0.1$ and $\Delta t = 5/60\,\mathrm{h}$, a $500\,\$/\mathrm{MWh}$ RT price spike produces a $\approx 417\,\mathrm{kW}$ response — large enough to be clearly visible while remaining within the PCC cap. A larger γ (e.g. $\gamma = 1$) would anchor agents so tightly to their DA schedules that RT variation disappears into numerical noise; a smaller γ risks violating the PCC constraint on every step.

9 Findings

What the case study shows

Finding 1 — Equilibrium correctness. FACET-mpGNE produces the same Nash equilibrium as the iterative ADMM baseline. Total cost agrees to within 0.66% (Table 4); per-agent costs agree to within ±2%. The power trajectories (Fig. 11) and state trajectories (Fig. 12) overlap visually for the full week. This validates the explicit-map approach as an exact replacement for ADMM, not an approximation.

Finding 2 — Iteration-free 98.9% of the time. On 1424 of 1440 RT steps (98.9%), FACET resolves the GNE via direct CR lookup with **zero** communication rounds (Fig. 8 right). Only 16 steps (1.1%) trigger an ADMM fallback. ADMM uses 183,112 total rounds (median 125/step, p95 193) versus FACET's ≈2,000 fallback rounds — a **92**× reduction (Fig. 9, Table 3).

Finding 3 — Data-transfer reduction is $\sim 50\times$. Total floats transferred over the week: 4,394,688 (ADMM) vs 87,872 (FACET), a **50**× reduction (Table 3).

Finding 4 — Collapsing DA to a 1-scalar parameter halves the θ dimension. Reducing the DA reference from $H = 6$ per-step values to a single hourly scalar cuts $|\theta_i|$ from 19 to 14 (25 to 20 for PV+Batt), giving 9,242–11,334 total CRs per day (Table 2) — a significant reduction from the 60,000+ CRs the original full-horizon formulation would require. The mpQP is still rebuilt only once per day.

Finding 5 — Two-timescale decomposition is the right architecture. By keeping the DAM solve centralized (it's a one-shot daily computation handled by the ISO anyway) and pushing only the RT GNE into distributed execution, we obtain both the policy benefits of centralized planning and the latency/communication benefits of distributed control. The 5-min RT loop is the time-critical part where iteration-free GNE matters.

10 Conclusion

The case study demonstrates that an explicit multi-parametric Generalized Nash Equilibrium map can be deployed in an autonomous market-participation setting at the 5-minute real-time cadence without iterative coordination. Two design choices make this possible: (i) dropping the global lower coupling $L_{\min}$ which is non-binding in practice but doubles the coupling rows, and (ii) rebuilding the mpQP *daily* with a tight DA box derived from the actual DAM clear, so the parametric volume stays narrow regardless of the agents' nominal bidding ranges. The resulting FACET-mpGNE matches the equilibrium produced by an ADMM baseline to within 2% on every agent while reducing communication rounds by 92× and floating-point traffic by $\sim 50\times$.

For a 4-agent, $H = 6$ horizon problem on real PJM-RTO data, the offline daily mpQP rebuild takes 40–65 s and the online evaluation completes in $\approx$ 200 ms per RT step (98.9% of which is point location in the explicit map; the remaining 1.1% triggers an ADMM fallback). Reducing the DA parameter from $H = 6$ values to a single hourly scalar cuts $|\theta_i|$ from 19 to 14, keeping total CRs under 12,000 per day while eliminating any per-agent bidding-range constraint from the formulation. The pipeline is ready to extend to longer horizons, more agents, or richer agent dynamics by the same per-day-box mechanism.